

Task 3 - One-Page Summary

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Design choice in plain language

This project designs and partially implements a jurisdiction-aware AI credit governance tool for Citi, comparing US fair-lending explainability expectations with EU AI Act high-risk AI governance expectations. The prototype uses 5,000 rows of synthetic credit application data, a logistic-regression credit risk model, configurable jurisdiction rules and portfolio-level fairness monitoring. The same portfolio produces an approval parity ratio of 0.885. In US mode, the tool compares this result with a 0.80 threshold and emphasizes adverse-action reasons and fair-lending review. In EU mode, the same result is compared with Meridian's internal 0.90 governance threshold and triggers high-risk AI governance actions, including bias testing, documentation and human oversight. The threshold is not an AI Act statutory number; Article 10, 14 and 17 obligations are mapped separately. The demonstration model achieved AUC 0.800 and accuracy 0.876 on a held-out synthetic test set. The tool also reports false positive and false negative rates by group and includes a sensitivity analysis showing how small score-distribution shifts can change escalation conclusions. It is intentionally limited: it is not legal advice, not a production credit approval engine, not trained on Citi customer data and not a replacement for human judgement. Its value is that it turns regulatory divergence into executable governance logic rather than a static memo.

What senior management should take away

The product is right for a cross-border bank because it helps compliance, risk and AI governance teams see the same model through different regulatory lenses. It solves a problem a spreadsheet cannot solve well: rule versioning, repeated metric calculation, jurisdiction-specific reporting and explicit escalation actions. Its most important limitation is that it demonstrates governance logic with synthetic data; production use would require validated customer data, legal review, model validation and operational controls.